



No. of Functional Features Included in the Solution

Date	10 JUNE 2026
Team ID	XXXXXX
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Audio Recording & Ingestion	Captures raw voice explanations via browser microphone inputs or multiformat media uploads.	audio_utils.py / Frontend	DONE	1
2	Acoustic Feature Isolation	Extracts core signal waveforms, detects silent segments, and evaluates overall speech pause thresholds.	audio_utils.py / Backend	DONE	1
3	AI Speech-to-Text Conversion	Utilizes the OpenAI Whisper model pipeline to map speech arrays into structured, time-synced written text records.	speech_to_text.py	DONE	1

4	Semantic Similarity Check	Implements Sentence-BERT context embeddings to evaluate thematic alignment margins against master references.	<code>semantic_eval.py</code>	DONE	0.5
5	Speech Flaw Density Profiler	Parses generated transcripts to quantify delivery stutter tokens, phrase breakdowns, and filler-word patterns.	<code>text_utils.py</code>	DONE	0.5
6	Metrics Dashboard Engine	Aggregates mathematical performance vectors into interactive chart visuals and understanding bands.	<code>scoring_engine.py</code> / Frontend	DONE	0.5
7	Automated Evaluation Reporter	Compiles detailed student transcripts, analytics tables, and charts into a downloadable, fixed-format PDF summary.	<code>report_generator.py</code>	DONE	0.5
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S.No	Category	Features in Category	Example Features
1	User Interface (UI)	Interactive data components, media upload arrays, audio widgets, and charts.	Streamlit dynamic dashboard view, micro-recorder controls.
2	Backend / Logic	Acoustic feature engineering, automated score mapping, and speech flaw profiling logic.	Librosa pause tracking loops, text word-frequency counters.

3	Database / Storage	Input storage queues and secure delivery endpoints for generated analytical files.	Local system directories for volatile uploads and generated PDFs
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Feature Summary:

Metric	Count / Value
Total Features Planned	7
Total Features Implemented	7
Core / Must-Have Features	5 (Audio Ingestion, Transcription, Semantic Check, Speech Profiler, Dashboard)

Additional / Nice-to-Have Features	2 (Acoustic Feature Isolation, Automated PDF Report Generation)
Features Tested & Verified	7

Feature Category Breakdown:

4	API / Integration	Advanced machine learning pipeline connectors and text mapping systems	OpenAI Whisper translation engine, Sentence-BERT transformer models.
5	Security / Authentication	Basic localized folder isolation and storage clearance protocols.	ession-state route controllers and automated /temp/ directory cache flushes.